



The YRB annually generates a GDP of over US\$4.4 trillion (25 percent of the China's GDP and nearly 5 percent of the world's economy) and produces more than one-third of the total national grain and meat.

Estimates by the Ministry of Water Resources (MWR) suggest that the river's natural annual flows have decreased significantly, with the 2001-2015 annual average being 25 percent lower than that in 1956-1979.<sup>9</sup> In addition, the water resources development and utilization rate – the ratio of total water use to total water resources in a river basin or region – has reached 80 percent, significantly exceeding the nationally determined sustainable utilization threshold of 40 percent.<sup>10</sup>

Mo, Xing-Guo, Shi Hua, Zhong-Hui Lin, Su-Xia Liu, and Jun Xia.



The World Bank-supported Yellow River Basin Ecological Protection and Environmental Pollution Control Program (P172806) invests in the middle reaches of the YRB in Henan and Shaanxi provinces (hereafter referred to as YRB-HS), and has water scarcity as its primary focus.

The water balance serves as the foundation for identifying and prioritizing sectoral interventions so that water demand for development does not exceed the water thresholds and encourages and incentivizes approaches and investments that optimize water efficiency.

Gansu province had the lowest GDP per capita in China at RMB 41,046 yuan (about US\$5,780) in 2021, less than one quarter of that of Beijing.

Investments in nature-based solutions, such as riparian woodlands, can improve both water storage and flood resilience.



As in the case of YRB-HS, the PforR instrument was also selected for this YRB-GS Program as it can leverage significant resources under existing Government programs to achieve impacts beyond traditional Investment Project Financing (IPF) operation.

Guided by the YRB Plan, Gansu and Shandong developed their respective provincial Yellow River Basin Ecological Protection and High-quality Development Plan (hereafter the Gansu YRB Plan and the Shandong YRB Plan) for 2021 - 2030.

The proposed PforR (·the Program·) is intended to support institutional interventions at the provincial level and direct investments for ecosystem restoration and protection in selected watersheds in Gansu and Shandong Provinces.













The proposed YRB-GS will support Gansu and Shandong in implementing their provincial and local investment plans to achieve the ecosystem restoration and protection objective specified in the YRB Plan and the Yellow River Protection Law.

The Program activities include physical investments in the six municipalities and technical assistance in institutional and capacity strengthening at the provincial and municipal/county levels.







The assessment compared a no Program scenario to a scenario of a government program that includes World Bank support.<sup>2</sup> 0 For forest activities, the Program economic rates of return (ERRs) with and without GHG emission reduction are estimated at 13 percent and 16 percent, respectively, i.e., significantly above the discount rate of six percent.<sup>2</sup> 1 Detailed comparisons of alternatives and selected options for the rehabilitation of waste water treatment plants and sewage collection systems, as well as wetland improvement, will be conducted when the feasibility study reports are prepared for location specific interventions by the county development and reform committees.

<sup>2</sup> 1 The discount rate of 6 percent is recommended for investments with long-term unquantified social and environmental benefits. Technical Note on Discounting Costs and Benefits in Economic Analysis of World Bank Projects; and NDRC.

An overall assessment of the legal framework and the procurement system found them to meet the Bank's principles of economy, efficiency, effectiveness, transparency, and accountability.





























Monitors both upgraded  
and newly built township  
wastewater treatment  
plants in Program  
municipalities and  
counties.



































- Technical standard for saline-alkaline land amelioration in landscape project (CJJT283-2018)
- Technical specification for soil salinity treatment in landscape projects (CJJ/T283-2018) ·
- Technical guidelines for ecologically-clean small watershed construction (SL534-2013)
- Technical specification for soil salinity treatment in coastal areas (DB32/T 4313-2022)

Disbursement will be made scalable based on the number of hectares (0.01 hectare) implemented with eligible soil salinity treatment measures and will not be capped at the annual target within the total of 427 hectares within the Program implementation period.





















